SAI PRANAV ADABALA

Mechanical and Mechatronics Engineer.

Michelsdorf 5, 93413 Cham

(+49) 15563469340

pranav.infinite@gmail.com

www.linkedin.com/in/pranav21

https://github.com/asp21523

EDUCATION

03/2025-Present
Relevant courses:

- Cyber Physical Systems.
- Advanced modelling and simulation.
- Functional safety.
- Autonomous Systems.
- Advanced Robotics.

07/2019-05/2023
Relevant courses:

- Reliability, maintenance Engineering
- Metrology and QA
- AI in manufacturing.

About Me

Aspiring Mechatronics Engineer pursuing an M.Eng. in Mechatronics & Cyber-Physical Systems at Deggendorf Institute of Technology, with a background in Mechanical Engineering and hands-on experience in automotive manufacturing. Skilled in GD&T, Body-in-White quality assurance, PFMEA, and process error-proofing, with a strong analytical approach to quality improvement. Interested in quality data analysis, FMEA, and AI-driven quality methods, and eager to contribute to advanced automotive technologies.

M.Eng. Mechatronics & Cyber Physical Systems

Deggendorf Institute of Technology, Campus Cham, Current Grade: 1,2

- Functional Safety – Fault Tree Analysis (FTA), reliability modelling, risk assessment based on IEC 61508, Safety Integrity Levels (SIL), probabilistic reliability evaluation.
- Cyber-Physical Systems – requirements engineering, system design, product lifecycle management, risk analysis (D-FMEA), safety-oriented development.
- Advanced Modelling & Simulation – dynamic system modelling, system identification, parameter estimation, state-space modelling and simulation tools.

B.Tech. Mechanical Engineering

Visvesvaraya National Institute of Technology, Nagpur, India

- Secured a final grade of 8.19 CGPA.
- Bachelor’s Thesis: Electric Vehicle Battery Thermal Management System.
- Designed and analysed thermal behaviour of the battery pack with 18650 cells using SolidWorks and Ansys.
- A temperature drop up to 60% is achieved with the help of the serpentine duct.

WORK EXPERIENCE

07/2023-03/2024

Quality Engineer

TATA MOTORS PASSENGER VEHICLE Ltd, India.

- Coordinated a 20-member quality team for Body-in-White dimensional inspection and laser brazing validation.
- Supported Process FMEA development for the Tata Punch Electric Vehicle program, contributing to risk identification and failure prevention.
- Identified the root cause of a recurring production defect and implemented a Poka-Yoke error-proofing solution, reducing defect occurrence from 5 PPH to 0 PPH.

05/2022-06/2022

Project Intern

BHARAT HEAVY ELECTRICALS LIMITED, India

- Developed MATLAB code for gas turbine wheel balancing to reduce vibrations and imbalance in the assembly.
- Studied GD&T principles in gas turbine manufacturing, which ensures precise component alignment.

PROJECTS

03/2025-Current

- Welding Fault Prediction using Machine Learning
  - Developed a Random Forest model to predict welding defects using process parameters (current, voltage, force).
  - Performed data pre-processing, feature engineering, and model evaluation using precision, recall, F1-score, and confusion matrix.
  - Achieved 96.8% accuracy on unseen test data using stratified validation.
  - Identified voltage and current variability as key drivers of weld defects through feature importance analysis.
- PFMEA & Process Improvement – Rear Quarter Assembly Fixture
  - Conducted Process Flow Diagram (PFD) analysis and PFMEA for a rear quarter assembly fixture in BIW.
  - Identified a part mix-up failure mode (~5 PPH) caused by incorrect vehicle variant components entering the station.
  - Investigated the root cause and determined an incompatible bracket preventing proper fixture closure.
  - Implemented a capacitive sensor poka-yoke to detect incorrect components and prevent mis-assembly.

LANGUAGES

- English – Fluent.
- German – Intermediate.

TECHNICAL SKILLS

- Data Analysis & AI
  - Python (pandas, scikit-learn)
  - MATLAB
  - Microsoft Excel
- Programming
  - Python
  - C++
- Engineering & Simulation
  - SolidWorks
  - Fusion 360
  - ANSYS
- Robotics & Systems
  - ROS Noetic (basic knowledge)

SOFT SKILLS

- Initiative and problem solving mindset.
- Cross-functional Collaboration.
- Adaptability in grasping new tools and technologies.

CERTIFICATIONS

- AutoCad Deign, MSME-Tool Room, Hyderabad, 27/07/2021.
- CNC Programming, MSME-Tool Room, Hyderabad, 27/07/2021.

INTERESTS

- Quality Data Analysis
- AI Applications in Manufacturing
- Reliability Engineering & Risk Analysis
- Automotive Systems & Driving Technologies

Cham, 5th March 2026

A·Sai Pranav.